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2019, ICRTAC 2019Image Dehazing using Improved Dark Channel Prior and Relativity
of Gaussian

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Abstract

Bad climatic weather conditions occur in our daily life are snow, sandstorm, mist, haze and fog etc. These various atmospheric particles affect the natural scene and the visibility will be tampered. Different dehazing methods are used to restore the visual perception of the image, increase the natural panoramic view and preserve the structural features of the image. Poor visibility becomes a significant issue for most outdoor vision applications. Image dehazing is a highlighted research area because of its real time applications in surveillance systems, driver assistance system especially for the people who resides in hilly areas where mist and haze is prominent. This paper presents an modified dark channel prior based image dehazing system with an enhanced and refined transmission map constructed by utilizing the RGB and YCrCb color system. Transmission map is constructed and estimated from combined airlight component obtained from the above said color models. In this work, we compute the Relativity of Gaussian for RGB and YCbCr color space to produce guidance image. The both guidance images thus obtained are applied to guided filter for smoothening. Finally, hazeless image reconstruction is performed by utilizing the guidance images obtained from the previous step. In this proposed work, we focus on improving the transmission map by utilizing the concept of Relativity of Gaussian and Guided filter. We can find that the proposed method does not over enhance the images. Results analysis focuses on investigating the efficiency of proposed method by reconstructing the image without distortion, with no color artifacts and better aesthetic visual image

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1. Introduction

Environmental impact such as haze and fog have an unfavorable effect on the image quality. It will be very difficult and challenging to use such images in driver assistance systems, monitoring surveillance systems and outdoor and indoor object detection and recognition which require clear visible and high quality image for further processing. Usually, restoring the degraded hazy images from single input image is pretty challenging and tedious. Haze or fog is a prevalent natural phenomenon created by atmosphere particles that are suspended. Therefore, haze removal is a significant task to restore the visibility in images. There are many problems that need to be addressed in haze removal such as efficient computation of atmospheric air light and transmission depth map. Improved Dark Channel Prior (DCP)[4] is used to improve the transmission depth map construction in our work. The contribution that we introduce focuses on a considerable betterment of the reconstruction module. DCP-based image dehazing technique[2][4] is used to prevent blocking artifacts with enhanced transmission map. It calculates the three transmission maps based on each R, G, B components and one transmission map based on the luminance component of the YCrCb color system. Two images are reconstructed after applying guided filter in transmission map. In this paper, Relativity of Gaussian (RoG)[7] and Guided filter[6] is applied to enhance the transmission depth map for removing the existence of haze.

The rest of paper is arranged as following sections. Section 2 outline about the dehazing given in literature. Section 3 deals with atmospheric scattering model. Section 4 illustrates the detailed working of proposed method. Single Image Dehazing based on Improved Dark Channel Prior and Relativity of Gaussian (RoG) method, Section 5 and 6 gives an results and assessment metrics of the proposed method, Section 7 deals with the conclusion and the summarizes the contributions.

2. Literature Review

This section provides a comprehensive overview of distinct techniques for removing single image haze. Extensive research has been conducted in the field of haze detection and visibility improvement in hazy scenarios. We have single image haze removal and multiple image haze elimination techniques in the literature. In the, work we concentrate on eliminating haze from a single image. Purging haze from multiple images is time consuming and computationally intensive. Elimination of haze from single image is simple and easy to manipulate. Currently most of the haze removal tasks are done using CNN based techniques. Ren *et al* [12] proposed a deep learning approach on multi-scale CNN (MSCNN) technique for the haze elimination. Cai *et al* [13] investigated a deep CNN technique DehazeNet which significantly improves the haze elimination process thereby obtaining better hazeless images. A brief review is shown in Table 1.

Table 1. Literature Review

Author	Dehazing based on	Method	Advantage
Salazar et.al[3]	Algorithm using Morphological Operations	Morphological Operations used to identify the features of shape of object	Fast Computing of transmission depth map
Kaiming He et.al[6]	Using Dark channel prior	Dark channel prior is based on the estimation of dark pixels which helps in computing transmission of haze.	Construct transmission depth map directly from hazy outdoor images
Tufail et.al[4]	Improved Dark Channel Prior from color models RGB and YCbCr .	Transmission maps are computed for RGB channels and luminance channel in YCbCr	Better scene visibility
Tsung et.al[5]	Airlight Correction and filters applied	Airlight White Correction performed and filters based on local light and prespective prior applied	Better visual quality, helps in removing color casts.

In existing method[4], the technique of image dehazing is discussed based on the dark channel (DCP)[2]. Existing image dehazing techniques using traditional DCP can be improved or enhanced for image dehazing task. The experimental result shows that the existing method produces over contrast and edges are not properly identified. In such case, a method is introduced which rise above the inadequacy of the existing method.

3. Atmospheric Scattering Model

In atmospheric scattering model, when an image is captured during day time, scattering of light will occur in the environment. When the illumination of the dispersed light is scattered in various directions towards the viewer (camera) resulting in an additive radiance. This concept is modeled mathematically as:

$$I(x) = J(X)t(x) + A(1 - t(x)) \quad (1)$$

I is Observed Intensity of hazy image, J represents scene brightness, A describes air light and variable t represents transmission media that describe the part of the light rise that has not been distributed or scattered that is captured by the sensor of the camera. The term $J(x)d(x)$ represent as the attenuation. Direct attenuation depicts the radiance of the scene and is decaying in the medium. The equation for computing the transmission depth map t is given.

$$t(x) = e^{-\beta d(x)} \quad (2)$$

β is the Scattering Coefficient of the atmosphere. For the image is taken in the clear weather $\beta \approx 0$, its effect is neglected and we get $I=J$. When image is taken in hazy weather, the $\beta > 0$ and it becomes non negligible.

4. Proposed Dehazing Method based on Improved Dark Channel Prior and Relativity of Gaussian

The proposed method concentrates on detailed enhancement of the transmission map and hence reconstructing the hazeless image. Existing method [4] suffers from lack of proper reconstruction of the haze free image. To solve these issues, RoG is applied on the guided filter and improved dark channel prior is computed on the hazy image in our work. The existing method only performs the refinement of transmission map using laplacian filter and the images contain excess contrast and some edges are not properly identified. The DCP is calculated by taking the advantages of both the color system i.e, RGB and Y channel of YCbCr color system. The influence of the airlight can be computed with the concept of Dark channel prior estimation. After the estimation of air light, compute the transmission map using air light from RGB channels and air light from YCbCr, it can be denoted as A and A_y respectively. Transmission map enhancement is done using RoG [7][8]. We compute the RoG for RGB and YCbCr color space to produce guidance image. Both guidance images are added to a guided filter for smoothening to generate $t'(x)$ and $t'_y(x)$. In the reconstruction step, two images thus obtained are incorporated to restore and reconstruct the hazeless image. A new technique is introduced in this work to enhance the transmission depth map and used Relativity of Gaussian (RoG) and Guided filter for detailed enhancement in the no haze images.

The modules are given as follows and it is portrayed in Fig1.

- RGB to YCbCr Conversion
- Air light Computation
- Transmission map Estimation
- Transmission map Enhancement using RoG and Guided filter

4.1. RGB to YCbCr Conversion

In this work, we exploit the possibilities of using RGB and YCbCr images for the construction of dark channel prior. So as the initial step input RGB is converted to YCbCr. We take both RGB and the luminance(Y) image of YCbCr image for computing the improved Dark channel Prior for our work.

4.2 Dark Channel Estimation

In this work, the enhanced DCP [2][4] is computed on RGB color space and YCbCr color space. We need to compute the minimum dark channel prior. In the hazy image, dark channel prior is implemented to assess the thickness or amount of haze and to compute the haze-free image. In this work, the above said dark channel prior is constructed from YCbCr and RGB color system. 31×31 is the window size used as filter to obtain the previous dark channel. The minimum filter is applied on the filtered RGB image for three color components and also applied on the local image patch. The tiny patch size is not used because it provides erroneous assessment of air light, so we use large patch size.

The mathematical expression of dark channel prior on the RGB is given as:-

$$I_{dark} = \min_{y \in \Omega(x)} (\min_{c \in \{R,G,B\}} (I^c)) \quad (3)$$

$\Omega=31$ for the local patch size. The mathematical expression of dark channel prior on the YCbCr is given as:-

$$I_{dark-y} = \min_{y \in \Omega(x)} (I^c) \quad (4)$$

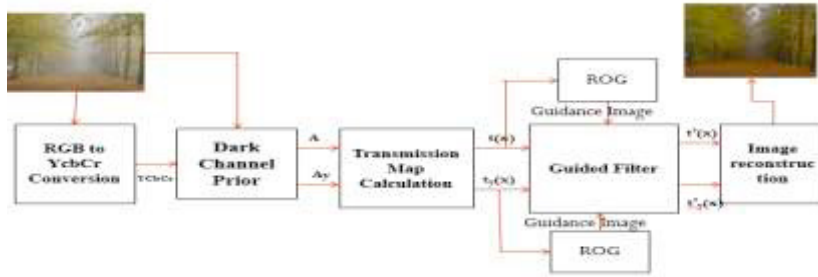


Fig 1: Image Dehazing using DCP & RoG

4.3 Airlight Computation

Air light computation[1] is done without computing dense haze area. The value of the atmospheric light is computed based on the depth map computation. After obtaining the Dark channel prior from the RGB and Y channel of YCbCr color system. Air light A and A_y are evaluated from dark channels pixels I_{dark} and $I_{dark,y}$ from RGB and YCbCr color models in the order. The coordinates of the bright top 0.1% pixels are identified and the maximal intensity values are taken as air light. The vector of A is 3×1 , where individual values correspond to maximum pixel intensity value in the each R, G, B channel. The above said steps are similarly applied in $I_{dark,y}$ to produce A_y [4].

4.4 Transmission Map Estimation

Here in this paper, the transmission depth map is used to represent the transmission map of a hazy image. The transmission depth map[4] represents the light rays that is not dispersed or scattered when captured in sensor. Similar to airlight computation, transmission depth map is constructed from both the RGB and Y channel of YCbCr color systems. The transmission map for RGB is denoted as $t(x)$ and transmission map of Y channel is denoted as $t_y(x)$. The transmission depth map for both the color space is calculated by the equation given below.

$$t(x) = 1 - \omega \frac{1}{3} \left(\sum_{c=1}^3 \left(\frac{I_c^e}{A_c} \right) \right) \quad (5)$$

$\omega=0.6\%$, amount of haze we are considering.

$$t_y(x) = 1 - \omega \left(\frac{I_y}{\max(A_y)} \right) \quad (6)$$

4.5 Transmission Map Enhancement

In proposed system, we use guided filter and Relativity of Gaussian for enhancement of transmission map.

4.5.1 Relativity of Gaussian (RoG)

Relativity of Gaussian can be applied to hazy images for detail enhancement [8]. We adopt detail improvement based on Relativity of Gaussian (RoG)[7] to boost the structure and details within the image while preserving the edges. RoG is applied to the reconstructed Y channel and a smoothened result is achieved. Relativity of Gaussian is utilized for the transmission map $t(x)$ and $t_y(x)$. Guided filter is applied to the transmission map where guidance image is the Relativity of Gaussian map is obtained. RoG based detail enhancement is to improve the detail within the image while preserving the edges. Edge preserving smoothening can be split in to two: Local filter and Global optimization [7]. Local filter will introduce Gibbs phenomenon resulting ringing and staircase distortions on edges. Global optimization focus on comparatively small variance suppression. Local regularization is called RoG that is optimized to define the corners on a selective scale. In our work, σ_1 and σ_2 are two parameters σ_1 is set as 10 and σ_2 is set as 20.

$$R = \left| \frac{G_{\sigma_1, \nabla S}}{G_{\sigma_2, \nabla S}} \right| \text{ s.t. } \sigma_1 < \sigma_2 \quad (7)$$

4.5.2. Guided Filter

The guided or directed filter has edge preservation and gradient preservation [7] capabilities. We made use of the linear local model for the estimation of the guided image. Guided filter is commonly used for dehazed application especially for preserving the details and structure present in the image. In the existing system[4], Laplacian filter is implemented for the enhancement of transmission map, output gradients are deducted from the initial transmission map mean filter is implemented for smoothening. In our work, we used a Guided filter for smoothening and detail enhancement of the image.

4.6. Reconstruction of Haze Free Image

In final stage after computing all the parameters is to reconstruct the improved quality image with minimized haze influence. The image reconstruction process is given by,

$$\mathbf{J}(\mathbf{x}) = \frac{I-A}{\max(t(\mathbf{x}), t_0)} + A \quad (8)$$

$$J_y(\mathbf{x}) = \frac{I-A_y}{\max(t_y(\mathbf{x}), t_0)} + A_y \quad (9)$$

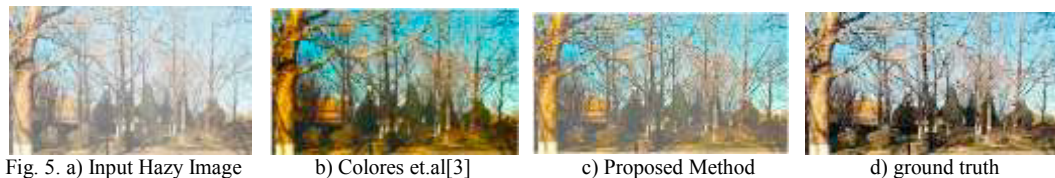
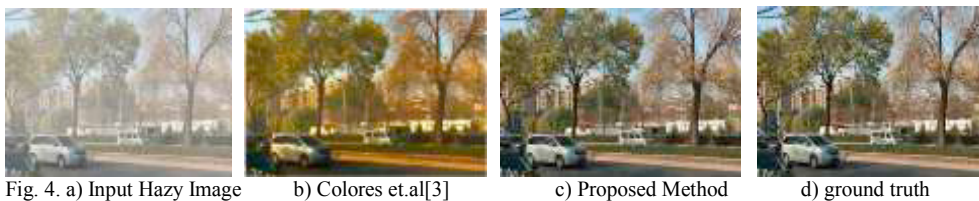
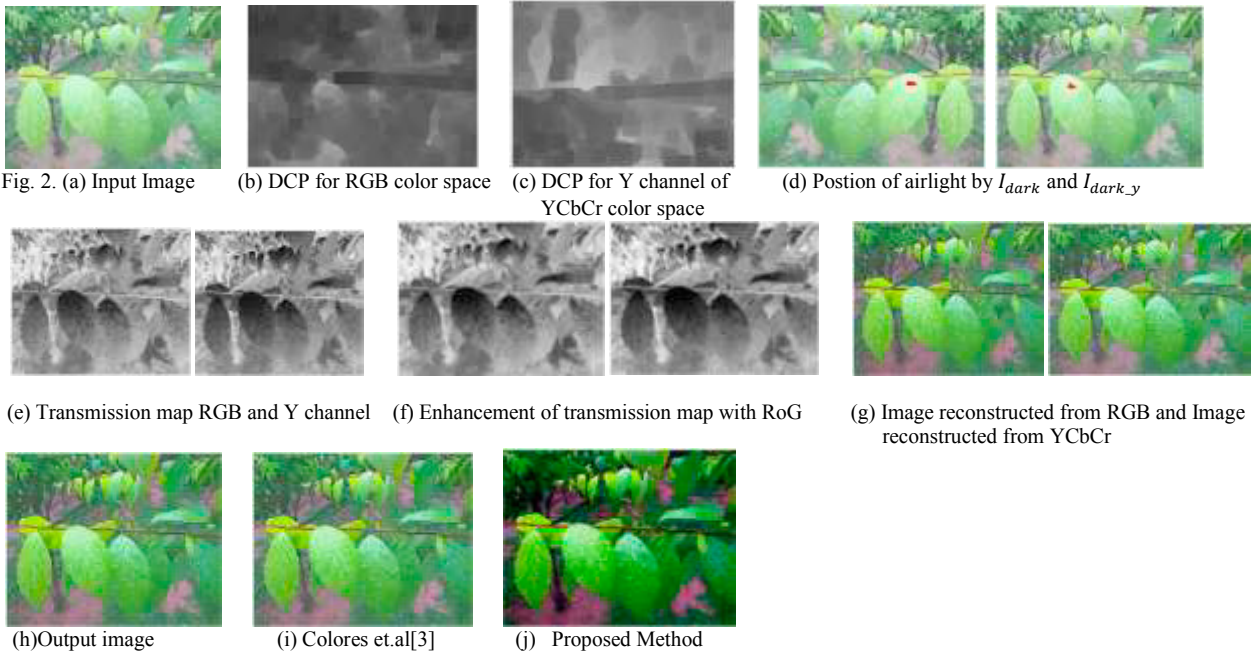
where t_0 is a constant, $\mathbf{J}(\mathbf{x})$ and $J_y(\mathbf{x})$ is the image reconstructed using the color space of RGB and is the image reconstruction using the color space of YCbCr[4].

$$\mathbf{R}(\mathbf{x}) = \alpha \mathbf{J}(\mathbf{x}) + (1 - \alpha) J_y(\mathbf{x}) \quad (10)$$

To combine $\mathbf{J}(\mathbf{x})$ and $J_y(\mathbf{x})$ images, a parameter α is used as weight. $\mathbf{R}(\mathbf{x})$ is the final reconstructed image. The value of α is set to 0.5.

5. Result Analysis

5.1 Qualitative Analysis



Colores et.al[3] result is haze free but it's color contrast is high and edges are not clear as in the reference image. Images obtained from the proposed method. From the qualitative assessment, it is clear that the result of the proposed method appear to be realistic ,visually pleasing and almost similar to the ground truth image

5.2 . Quantitative Analysis

The different evaluation measures employed to compute the performance of the proposed algorithm are SSIM, PSNR, BRISQUE and NIQE. We have performed the quantitative analysis in two ways. In the first case, test is conducted with the images where reference images or ground truth images are available and in the second case, where the reference images are not known or available. The metrics SSIM and PSNR are computed for the hazy images where reference image is available whereas BRISQUE and NIQE is computed for the images where ground truth images are not present .In both the above said cases, proposed technique outperform the existing method.

5.2.1 SSIM and PSNR (evaluation measure used when ground truth images are present)

The SSIM[9] (**Structural Similarity Index**) is a technique for assessing the similarity or closeness of the result image with the ground truth image. This is a good evaluation metric for identifying the structures and the edges present in the resultant output image. The range of the SSIM is from 0 to 1. **PSNR(Peak Signal to Noise Ratio)** denotes the quality of the reconstructed image by comparing with the ground truth of the image. PSNR is computed using the given equation and it is represented in decibals.

$$\text{PSNR} = 10 \log_{10} \left(\frac{\text{MAX}^2}{\text{MSE}} \right) \quad (11)$$

where MAX denote max intensity of input image and MSE is error.

Table 2. Quantitative Analysis Performance Evaluation

Image	Existing Method[4]		Proposed Method	
	SSIM	PSNR (dB)	SSIM	PSNR (dB)
Leaf	0.5719	35.9990	0.9149	40.9439
Building	0.5094	38.5463	0.8550	42.0662
City	0.5429	36.7623	0.8126	43.0335
River	0.6114	20.6935	0.8976	40.0294
Trees	0.7561	39.2957	0.9157	42.7431

4.2.3 BRISQUE and NIQE (evaluation measure used when ground truth images are not present or available)

BRISQUE (**Blind Reference less Image Spatial Quality Evaluator**) [10] is a evaluation measure usually used in image enhancement where the ground truth image is not known to the researchers. This metric is a quality for is computed based on the spatial feature. A lower score shows a better quality of perception. BRISQUE is a spatial domain based metric which computes the naturalness of an image based on the Image Quality Assessment (IQA). It is inspired from the concept of considering the adjacent luminance values and estimating the statistical products of those values. In this model, parameters are further quantified to estimate the naturalness of the image. **NIQE** (Naturalness Image Quality Evaluator) [11] is a evaluation measure computed based on the naturalness of an image. This metric is a no-reference image quality score. A lower score shows a better quality of perception. NIQE metric is inspired from the concept of computing statistical characteristics that depends on the Natural Scene Statistic (NSS). NIQE apply NSS features from patches of the image and fitting Multivariate Gaussian Model (MVG). The quality of the distorted image is expressed as

$$\mathbf{D}(\mathbf{v}_1, \mathbf{v}_2, \Sigma_1, \Sigma_2) = \sqrt{(\mathbf{v}_1 - \mathbf{v}_2)^T \left(\frac{\Sigma_1 + \Sigma_2}{2} \right)^{-1} (\mathbf{v}_1 - \mathbf{v}_2)} \quad (12)$$

where Σ_1, Σ_2 and $\mathbf{v}_1, \mathbf{v}_2$ denote the covariance and average of the natural MVG model between the original and distorted images. Without ground truth images, we have calculated performance evaluation using NIQE and BRISQUE. Table 2 describe quality analysis of existing method and proposed method using SSIM and PSNR. The highest value of SSIM and PSNR indicates better result. Table 3 assesses the naturalness of test images using BRISQUE and NIQE. The smaller BRISQUE and NIQE values, better the result.

Table 3. Quantitative Analysis Performance Evaluation

Image	Existing Method[4]		Proposed Method	
	BRISQUE	NIQE	BRISQUE	NIQE
Road	44.4183	28.6463	42.6199	23.1209
House	48.6075	20.9788	44.6079	15.6829
Cons	44.7045	28.1794	43.5352	20.5544
Forest	44.7045	22.9675	43.4604	16.3344

6. Conclusion

Usually the natural weather conditions such as haze or fog hinder the visibility and aesthetic view of a scene. An improved image dehazing system has been proposed exploring the concept of improved Dark Channel Prior (DCP) and Relativity of Guassian (RoG) with guided filter. Results show that the existing method produces over contrast, over enhancement and detailed information is hidden and is not properly identified. So this improved method calculates the transmission map and applied Relativity of Gaussian and Guided filter to enhance the transmission map. This work provides good performance for the task of obtaining hazeless image both qualitative and quantitative manner. Experimental results show that the proposed method estimates haze more accurately, the reconstruct the images are more realistic and detailed information is restored.

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